AUDIT BRIEF: COVERT WATERMARKING CAPACITY CONSTRAINTS

Abstract: Digital rights protection protocols often attempt to embed cryptographic recipient markers within compact document excerpts. This technical summary demonstrates the physical limits of text carrier capacity when embedding full untruncated 256-bit HMAC signatures.

Because a full cryptographic frame requires synchronization preambles, salt nonces, recipient identifiers, and integrity checksums, a minimum of approximately sixty lines of text is mathematically necessary. Documents possessing fewer words cannot host the security payload without truncating cryptographic margins. Consequently, high-assurance forensic systems must fail closed and report an explicit capacity deficit rather than silently degrading collision resistance.